
Winfried van den Dool

Amsterdam, The Netherlands
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Education

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| 2021 - present | Ph.D. in Machine Learning
QUVA Lab, University of Amsterdam, the Netherlands
Topic: Computational efficiency in deep learning, adaptive resource allocation for fast inference and sampling
Supervisor: Prof. dr. Max Welling |
| 2016 - 2019 | Master Mathematical Sciences (<i>cum laude</i>), Utrecht University
Specialization in probability, statistics and mathematical modeling
Thesis: A new method using Bayesian Neural Networks to generate explanations of Deep Learning decisions (<i>grade 9/10</i>)
Final GPA: 4.00 |
| 2016 - 2017 | Premaster Economics, Utrecht University
Final GPA: 4.00 |
| 2012 - 2016 | Bachelor Mathematics (<i>cum laude</i>), Utrecht University
Bachelor Physics and Astronomy (<i>cum laude</i>), Utrecht University
Minors in: Computer Science (<i>grade 9/10</i>), Econometrics (<i>grade 9/10</i>) and Philosophy of Science (honors program)
Thesis: A new algorithm for counting self-avoiding walks: the Tiling method (<i>9/10</i>)
Final GPA: 4.00 |

Work experience

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| 2025 | Research Internship (6 months) at Fermioniq on quantum algorithms for efficiently solving Partial Differential Equations using tensor network representations |
| 2018 - 2021 | Data Scientist at PricewaterhouseCoopers software developer focusing on the application of machine learning in finance, with an emphasis on explainable AI. |
| 2018 | Research Internship (6 months) at RaboBank doing causal inference research on macroeconomic variables and the influence of Quantitative Easing on yield curves. |

Publications

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| 2026 (Planned) | Efficient Long-Context Transformers via Hierarchical Vector Quantization
Winfried van den Dool, Patrick Forré, Amir Habibian, Yuki M. Asano, Max Welling |
| 2026 | <u>AVQ-Attention: Adaptive Vector-Quantized Attention</u> (ECCV)
Winfried van den Dool, Patrick Forré, Amir Habibian, Yuki M. Asano, Max Welling |
| 2025 | <u>Adaptive Mesh Quantization for Neural PDE Solvers</u> (TMLR, ICLR J2C certificate)
Winfried van den Dool, Maksim Zhdanov, Yuki M. Asano, Max Welling |
| 2023 | <u>Efficient Neural PDE-Solvers using Quantization Aware Training</u> (ICCV workshop)
Winfried van den Dool, Tijmen Blankevoort, Yuki M. Asano, Max Welling |